1. Hypothesis

Natural-language safety instructions can condition a learned residual flow model to modify a fixed base robot trajectory, satisfying different semantic safety constraints while preserving task success.

Specifically, for the same scene and base trajectory, changing only the safety instruction should produce an appropriate action correction.

$$\Delta A \sim p_\theta(\Delta A \mid s, A_{\text{base}}, E_{\text{text}}(T))$$

$$A_{\text{corrected}} = A_{\text{base}} + \Delta A$$

2. Experiment setup

Environment: Custom MuJoCo/LIBERO environment with a Franka arm, a mug, a laptop, and a placement target.

Task: Pick up the mug and place it at the target location.

Base policy: Start with a fixed, known pick-and-place trajectory. No VLA is required for the initial POC.

Safety instructions:

- **Constraint A:** "Avoid carrying the mug over the laptop."
- **Constraint B:** "Keep the mug upright to prevent spilling."
- **Constraint A+B:** Both constraints simultaneously.

Training:

- Generate paired nominal and safe trajectories for A and B separately using a controller or trajectory optimizer.
- Compute residual targets: $\Delta A^* = A_{\text{safe}} - A_{\text{base}}$.
- Encode safety instructions using a frozen text encoder (e.g., BGE).
- Train a conditional flow-matching model to generate residual action sequences.
- Do not train on the combined A+B instruction.

Evaluation metrics: Safety violation rate, task success, trajectory deviation from the base, and correction magnitude.

3. Three experiments

Experiment	Procedure	What it demonstrates
1. Text-conditioned action correction	Keep the scene and base trajectory fixed. Swap instruction A and B. Check whether A modifies the path to avoid the laptop and B modifies orientation to keep the mug upright. Compare against no-text and shuffled-text baselines.	Whether text causally influences the generated residual in the intended way.
2. Language generalization	Train using a few phrasings of A and B. Evaluate on unseen paraphrases, e.g., "Don't pass above the electronics" and "Maintain the cup's vertical orientation."	Whether the residual flow generalizes across different linguistic expressions of the same constraint.
3. Compositional generalization	Train on A and B independently, then test the unseen combined instruction A+B. Measure whether the corrected trajectory simultaneously avoids the laptop and keeps the mug upright while completing the task.	Whether the model can compose previously learned safety requirements without retraining.

Important experimental control: Ensure the base trajectory violates both A and B, and that separate feasible corrections exist. Also include varied scenes and trajectories so the model cannot simply memorize one correction for each text string.

Success criterion: The strongest result would be that changing the safety text changes the correction appropriately, unseen paraphrases preserve that behavior, and unseen combinations satisfy both constraints without sacrificing pick-and-place success.